Time Series Analysis: Session Learning Objectives

After this session, you will be able to

1. Recognise what is meant by a univariate time series.
2. Understand the difference between this and standard linear regression models.
3. Understand 1st/2nd Order Stationarity & methods to make series so.
4. Interpret (Partial) Auto- Correlation Functions (ACF) of various series.
5. Identify & interpret *Autoregressive* and *Moving average models.*
6. Examine ARMA/ ARIMA models to study suitable choice of coefficients.
7. Build ARMA/ ARIMA models for given datasets

The Story so Far...

- For some time series main focus is to forecast or simulate data.
  - If we can find structure in past data, it will likely help improve our forecasts and simulation accuracy.
- As seen, this structure can comprise trends & seasonality
  - Can decompose a series into these patterns.
  - However, even after this decomposition still have remainder component.
- Sometimes this remainder is made up of independent random variables.
  - i.e. Assume some underlying generating process for the time series.
  - This process is based on some statistical distribution from which variables are drawn.

The Story so Far... (/2)

- Frequently (e.g. in finance), consecutive elements of this remainder component are correlated[1])
  - e.g behaviour of sequential or nearby points in remainder affect each other in a dependent manner.
- The Auto-Correlation Function (ACF) is essential in characterising the remainder component to identify correlations' structure
  - Including AC structure in the model will improve the model's forecast
  - It will also improve realism of any simulated series based on the model.
  - Also helps in saying if a series is stationary or not!

[1] aka *serial correlation*.

Basics: Definitions and Notation

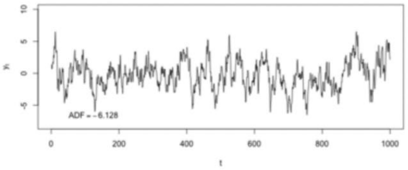
Univariate Time Series: a sequence of (scalar) measurements of the same variable over (usually regular intervals of) time.

- If data are equispaced, the time variable needn't be explicitly given as not used in the model itself.
- Usually assume in many time series techniques that data are *stationary*.
- This entails (as per the figure):

-> zero trend,

-> constant variance

-> const AC structure over time.



- If a time series is non-stationary, need to render it so.
- If we are unsure, need to examine and/ or apply a test.

Basics: Definitions and Notation (/2)

A Recap on Some Formal Definitions:

- *Expectation:* the arithmetic mean of random variable $E(x) = \mu$
- *Variance:* the expectation of the variable's squared deviations from the mean $\sigma^2(x) = E\left[(x-\mu)^2\right]$ (where $\sigma(x)$ is the standard deviation (SD))
- *Covariance:* measures how one variable x varies with a second, y $\sigma(x,y) = E\left[(x-\mu_x)(y-\mu_y)\right]$. ($\mu_x, \mu_y$: sample means of x, y resp).
- *Correlation:* scales the covariance from $[-1, 1]$: $\rho(x,y) = \sigma(x,y)/\sigma(x)\sigma(y)$. ($\sigma(x), \sigma(y)$ are sample SDs of x, y resp).
- *Autocorrelation*: time series correlation with a lagged version of itself

$$r_k = \frac{\Sigma_{t=k+1}^{T}(y_t - \bar{y})(y_{t-k} - \bar{y})}{\Sigma_{t=1}^{T}(y_t - \bar{y})^2}$$

- *Autocorrelation Function (ACF)*: A plot of correlation coefficients, r_k, between time series observations separated by k lags (y_t and y_{t-k}).

First & Second Order Stationary Series

First order stationarity:

A time series is first order stationary iff It is stationary in the mean (i.e. has constant mean).

Second order stationarity:

A time series is second order stationary iff It is stationary in the mean & variance (constant mean, variance, covariance).[a]

[a] Correlation between one value and next is only a function of the lag, i.e. the number of time steps separating each sequential observation.

Stationary or Non-Stationary?

- Time series with any repeated patterns (e.g. trends/ seasonality) are non-stationary
  – Both affect time series values at different t as both are functions of t.
- From Fig 2.1, a *white noise*[2] series is stationary but a *random walk* not.
  – RW is infinite sum of all past WN shocks so variance grows over time.
- Stationary time series have no long-term predictable patterns.

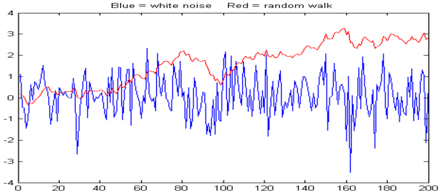


FIGURE 2.1: Random Walk Red and White Noise Blue

[2] One where little/ no autocorrelation outside certain limits

Stationary or Non-Stationary? (/2)

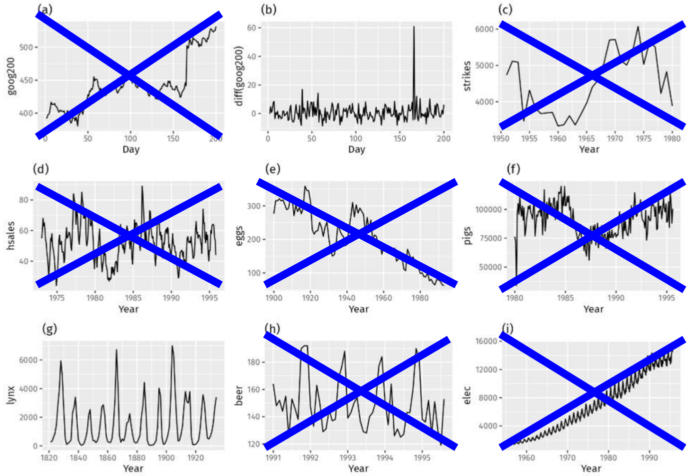


FIGURE 2.2: Various Diverse Time Series (from Hyndmann)

Stationary or Non-Stationary? (/3)

- Looking at Fig. 2.2 – Which are stationary (& why)?
  - (d), (h) and (i) are ruled out due to Seasonality.
  - (a), (c), (e), (f) & (i) due to trend & increasing level).
  - (i) also isn't a contender due to increasing Variance.
  - just remains (b) and (g) as stationary series.
- Strong cycles in series (g) seem to make it non-stationary.
- But these cycles are aperiodic
  - due to lynx population getting too big for available feed,
  - so breeding stops and the population falls very low,
  - then regenerating food sources lets numbers increase etc.
- Long-term, the timing of these cycles is not predictable. Hence the series is stationary.
- Better to carry out a unit-root test (e.g. Dickey-Fuller et al).

Ex. 2.2 A Stationarity Test: Augmented Dickey-Fuller

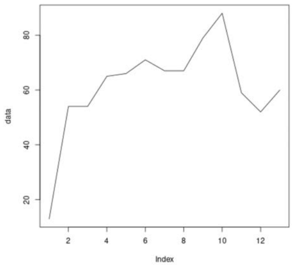


FIGURE 2.3: Plot of sample data

- Fig 2.3 shows a sample series
- R uses `adf.test` to test stationarity

```
library(tseries)
data <- c(13, 54, 54, 65, 66,
71, 67, 67, 79, 88, 59, 52, 60)
plot(data, type='l')
adf.test(data)
# Dickey-Fuller = -1.6549, Lag
order = 2, p-value = 0.7039
# alternative hypothesis:
stationary(data)
```

- Dickey-Fuller statistic should be -ve & $p < 0.05$ for stationarity
- $p > 0.05$ so non-stationary

Ex. 2.3 Another Stationarity Test: Lynx Data

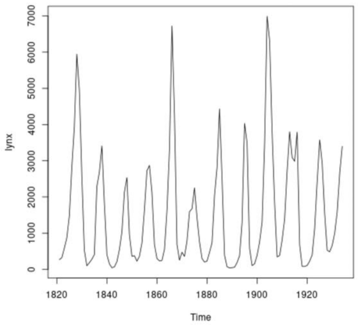


FIGURE 2.4: US Lynx Population Data 1820-1940

- Fig 2.4 shows Lynx Data in US
- Again, `adf.test` in `tseries` library to test stationarity

```
library(tseries)
library(fma)
plot(lynx, type='l')
adf.test(lynx)
# Dickey-Fuller = -6.3068,
Lag order = 4, p-value = 0.01
# alternative hypothesis:
stationary(data)
```

- $p < 0.05$ & DF statistic is negative so stationary

Examples of ACF Plots: Ex 2.4 (a) Random Values

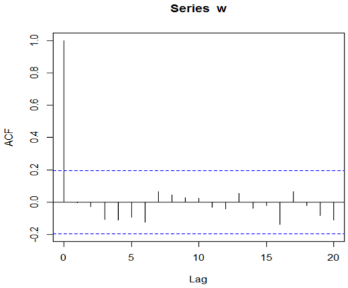
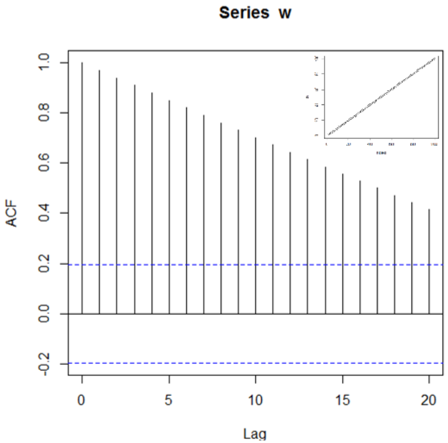


FIGURE 2.5: ACF of set of 100 randoms, w

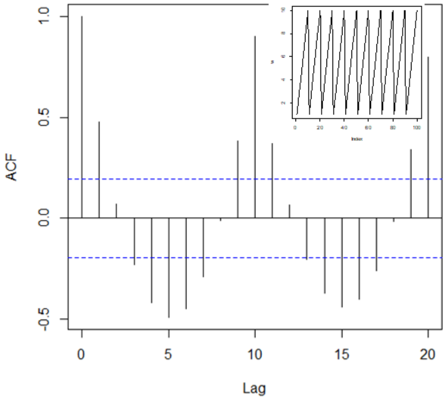
- ACF is useful to give series correlation structure
- Plot shows sequential values of lag $k = 0, 1, ..., n.$ for series of randoms
- Here, lag k autocorrelation is correlation between values k time periods apart.
- ACF says if model is well-fitted or if need to remove any more AC.
- Correlation at lag 0=1
- Blue lines are limits of null hypothesis that correlation at lag k is 0, at 5% level.
- Shows that series of randoms (w) is stationary

Examples of ACF Plots: Ex 2.4 (b) Fixed Trend



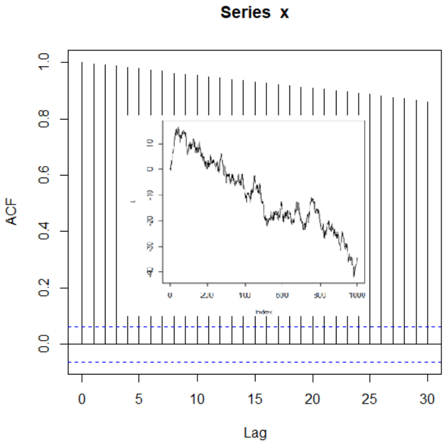
- Inset is series with linear trend.
- This model has ACF decreasing linearly with increasing lag.
- This is due to nearby observations having values of similar size
- -> AC at small lags tend to be large and +ve
- -> ACF like this is clear indication of a trend.
- This implies non-stationarity
- ? What would the ACF of a model with downward trend look like?

Examples of ACF Plots: Ex 2.4 (c) Repeated Sequence



- Periodic model shows peaks at lags $0, 10, 20$.
- -> First is perfect correlatiion of series with itself at lag 0
- -> Period is 10 so w aligns with itself perfectly at these points
- -> Amplitude of peaks decreases due to finite series length
- Negative (anti-correlation) peaks at lags 5 and 15 of exactly -0.5.
- -> Due to troughs at 5, 15 lags behind series peak
- Such an ACF shows seasonality not yet accounted for in a model.
- The significant peaks in ACF also imply non-stationarity

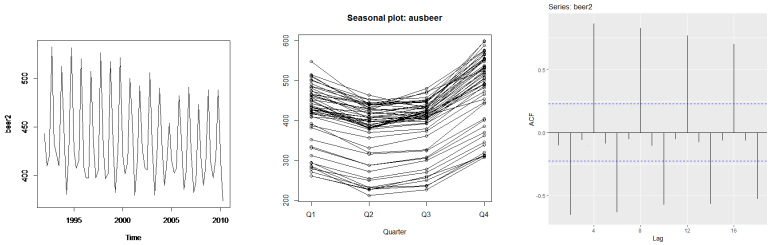
Examples of ACF Plots: Ex 2.4 (d) Random Walk



- RW model is where current value is previous plus random step up or down, see inset
- This is given by $x_t = x_{t-1} + w_t$ (w_t white noise)
- RW has zero mean[a] but time-varying variance, thus is non-stationary.

[a] Proof can be seen at `https://bookdown.org` at this link.

Examples of ACF Plots: Ex 2.5 Beer Again...



(a) Australian Beer Data (b) Seasonality per Year (c) ACF of Beer Data

FIGURE 2.6: Australian Beer Production Data from 1992 and ACF

- Australian Beer is shown in Fig 2.6(a), Seasonal in (b) and ACF in (c)
- Note seasonality in (c) and peaks at lag 4, 8, 12, 16 and at 2, 6, 10

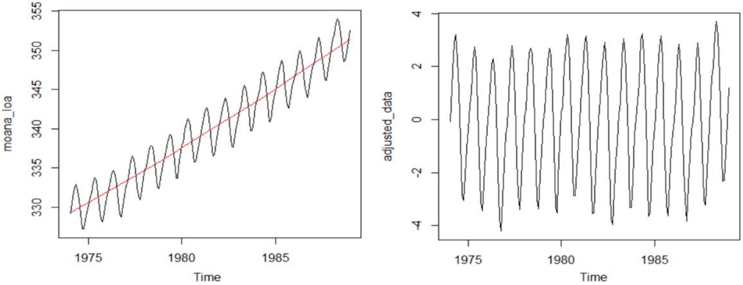
Non-Stationary Time Series: How to Remedy?

- Most series, even after SA, can still have trends, cycles, random-walking, and other non-stationary behaviour.
- If not stationary, time series can often be made stationary by either:

1. Rendering Difference-Stationary:
   - For stochastic trend, differencing gives stationary stochastic process, i.e., for x_t, create $Y_t = x_t - x_{t-1}$
   - Once usually enough. Too many times can *Over-Difference*.
   - Differenced Stationary Series Y_t has one less point than original.
2. Rendering Trend-Stationary:
   - With deterministic trend, can fit a curve to the data & model residuals as a residuals series.
   - Point is to remove long term trend, so use a simple fit, e.g. line.
   - Residual series is a stationary stochastic process.

Either way, stationary stochastic series is what is wanted

Ex 2.6: Removing a Deterministic Trend



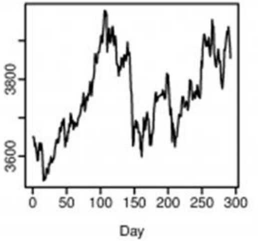
(a) Raw Data (b) Trend Removed

FIGURE 2.7: Mauna Loa CO2 Data

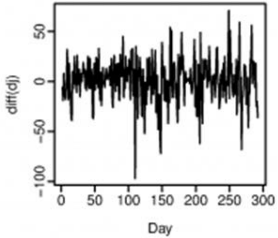
- Fig 2.7(a) shows CO2 raw data & Fig 2.7(b) with trend removed.
- Series now trend-stationary - (b) has approx constant level & variance.
- But residuals pattern shows data depart from model systematically.

-> Series is still non-stationary.

Ex 2.7: Rendering Difference Stationary



(a) Raw Data



(b) Differenced DJ Data

FIGURE 2.8: DJ Data

- 292 Days of DJ index (Fig 2.8(a)) & its Daily change in Fig 2.8(b)
- Which of the two graphs of related data is stationary?:
  - Increasing trend rules out stationarity for Fig 2.8(a).
  - Fig 2.8(b) has virtually zero trend and constant variance
  - Shows how differencing is used to render a series difference stationary

Ex 2.7: Rendering Difference Stationary (/2)

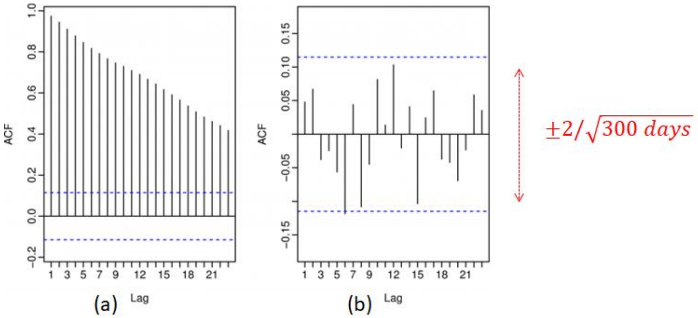


FIGURE 2.9: ACF of DJ and Differenced Data in Fig 2.8

- Interesting to look at ACF of Fig 2.8(a) & (b)
- Fig 2.8(a) ≈ random walk series - all AC lags outside limits $\frac{2}{\sqrt{T}}$
- Fig 2.8(b) ≈ white noise series - little/no AC outside limits $\frac{2}{\sqrt{T}}$
  - Suggests daily change is random & uncorrelated with past days.

Ex 2.7: Difference Stationarity: Over-Differencing (/3)

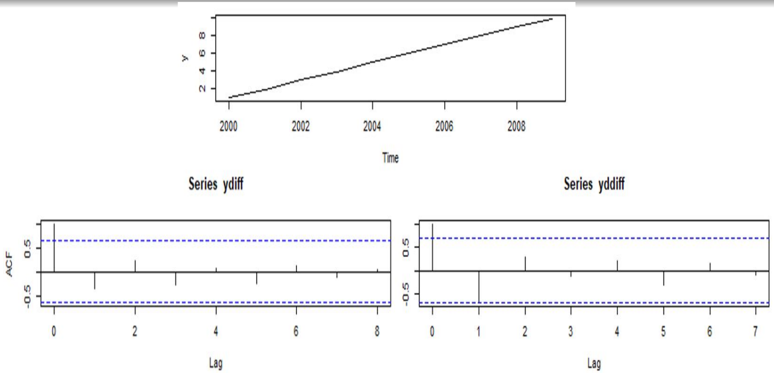


FIGURE 2.10: Fixed Trend with First & Second Differences

- Roughly fixed linear trend shown in Fig 2.10(a)
- First Diffs in (b) shows ≈ white noise - little/no AC outside limits
- Second Diffs in (c) - at least one lag outside limits

Model Types

Three basic types of *time domain* models:

1. Random Walk (RW)/White Noise (WN) Models
   - Can't really mimic frequency/ time behaviour of (eg.) financial data
2. Ordinary regression models using time indices as x-variables.
   - Basic model, underpins simple forecasting methods (as seen).
3. *Autoregressive Integrated Moving Average (ARIMA)* models
   - 3 parts in model: autoregressive (AR) & moving average (MA):
     - AR part regresses dependent variable on its past values.
     - MA part models the current series value as a linear combination of the current error term and those from the past
     - -> *Error* here represents a random shock or unexpected event
     - *Integrated* as it can involve differencing the series
   - We will look at these in more detail.

Model choice depends on trend, seasonality, stationarity etc.

Model Types 3: Auto-Regressive Models

- The First-order Auto-Regressive Model (AR(1))
  - First examine the theoretical properties of AR(1) model.
  - In it, x at time t is a linear function of x at $t-1$
  - Algebraically, can express the model as follows:

$$x_t \;=\; c + \phi_1 x_{t-1} + w_t \tag{2.1}$$

where c is constant,

x_t, x_{t-1} are the values of x at t, $t-1$

ϕ_1 is a parameter

w_t is a random variable representing the white noise part

- Assumptions of the First-order Auto-Regressive Model (AR(1)):
  - Errors $w_t \overset{iid}{\sim} N(0, \sigma_w^2)$
    i.e. errors are independently distributed with a normal distribution of mean 0 & constant variance)
  - The time series x_t is second-order stationary (implied by $|\phi_1 < 1|$)

Auto-Regressive Model: Ex 2.8(a) AR(1) & ACF

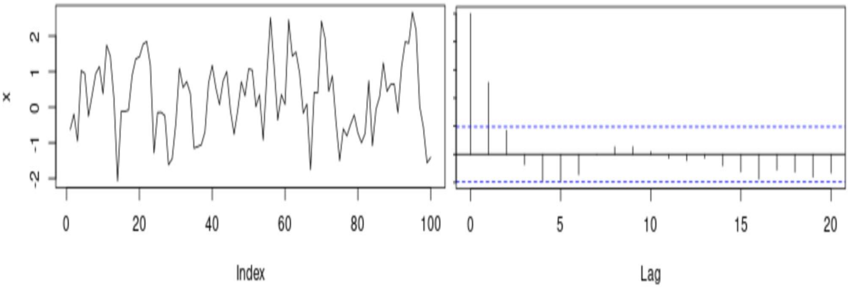


FIGURE 2.11: AR(1) & its ACF

- Top left plot in Fig 2.11 shows AR(1) Series $x_t = c + 0.6x_{t-1} + w_t$.
- To right, its ACF for first 20 lags. Note the tapering pattern.
  - For $+\phi_1$, ACF exponentially $\to 0$ as lag increases.
  - As expected, AR(1) has peak at $k = 1$, with tapering echoes in subsequent lags.

Auto-Regressive Model: Ex 2.8(b) AR(1) & ACF

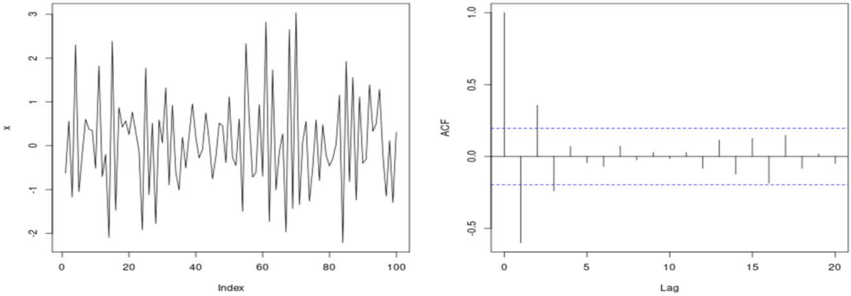


FIGURE 2.12: AR(1) with negative ϕ_1 & its ACF

- Top left plot in Fig 2.12 shows AR(1) Series $x_t = c - 0.6x_{t-1} + w_t$.
- To right, its ACF for first 20 lags. Note the oscillatory pattern.
  - For $-\phi_1$, ACF also exponentially $\to 0$ as lag increases
  - But the algebraic signs of the autocorrelations flip between $+$ & $-$
  - In AR(1) model, each lag is previous lag $\times$ the coefficient ϕ_1

Auto-Regressive Model: Ex 2.9(a) AR(2) & ACF

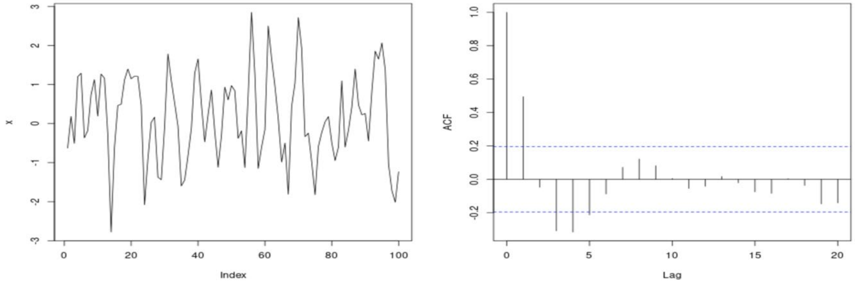


FIGURE 2.13: AR(2) & its ACF

- Fig 2.13 (top) has AR(2) Series $x_t = c + 0.66x_{t-1} - 0.33x_{t-2} + w_t$.
- ACF shows statistically significant peaks at $k = 1$, but also $k = 3, 4, 5$.
  - Lags 1, 2 followed by significant lags at 3, 4, 5, which are echoes from AR(2)'s internal momentum.

From math. decaying sinusoid due to complex z in $z^2 - \phi_1 z - \phi_2 = 0$

Reflections on Autocorrelation & ACF

- ACF shows linear r'ships between a value at t and at $t-1$, $t-2\ldots$.
- For AR(k) model, may wish to measure link between x_t & x_{t-k} *only* (i.e. omit influence of (i.e., $x_{t-1},\ldots,x_{t-(k-1)}$)
- To do this, need to transform the series to filter out $x_{t-1},\ldots,x_{t-(k-1)}$.
  - Calculate correlation of cleaned series: Partial Autocorrelation function (PACF).
  - Gives correlation between x_t & x_{t-k} with no spurious correlations (i.e. 'echoes').
  - AR(2): ACF in Fig 2.14(l) has multiple trailing lags, PACF (r) cuts off after lag 2

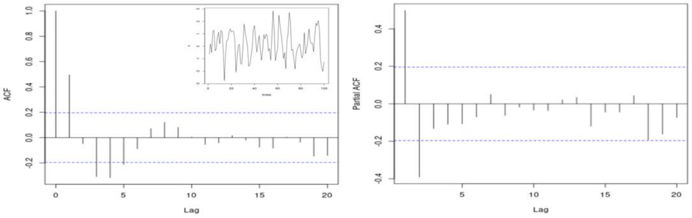


FIGURE 2.14: AR(2): ACF& PACF

Auto-Regressive Models: Ex 2.10 Earthquakes

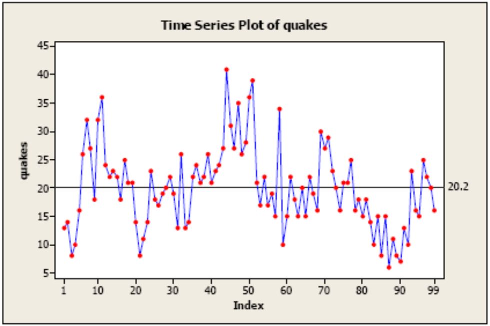


FIGURE 2.15: Global No. of Earthquakes of Magnitude ≥ 7.0 For 99 Years

- No firm trend in Fig 2.15: quakes = 20.2 shows series mean.
- Plot drifts either side of mean, remains briefly on each.
- There is no seasonality (annual data) and no obvious outliers.
- Hard to judge the variance.

Ex 2.10: Original and Lag(1) Series

t	x_t	x_{t-1} (lag 1)
1	13	–
2	14	13
3	8	14
4	10	8
5	16	10

TABLE 2.1: Quakes V Lag(1)

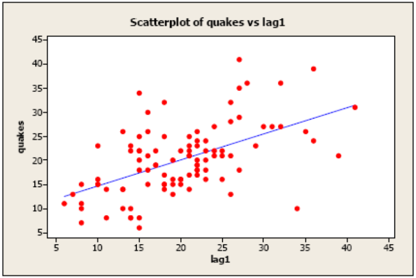


FIGURE 2.16: Quakes V Lag(1)

- Table 2.1 shows first 5 values with their lag 1 values
- Fig 2.16: Quake x_t V x_{t-1}
- Reasonable to plot V lag 1
- Only moderately related but there is a positive linear association so AR(1) model might be useful.

Ex 2.10: Residuals & ACF in an AR(1) Model

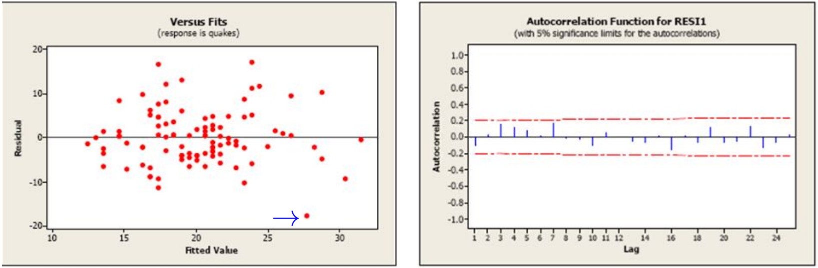


FIGURE 2.17: Residual and ACF of Fig. 2.15

- Fitting the data in Fig 2.16 to the AR(1) model Eqn.(2.1) gives the following relationship: Quakes, $x_t = 9.19 + 0.543x_{t-1}$
- Residuals plot Fig 2.17(L) doesn't show any serious problems (maybe an outlier at a fitted value of about 28?).
- Auto-Correlation function (R) seems ok too; all are within 95% lines.

Lecture Summary

In this session, we covered the following:

1. Grasping the meaning and properties of univariate time series.
2. Understanding 1st/2nd Order Stationarity & methods to make series so.
3. Following how observations relate to each other using (Auto)Correlation and the ACF.
4. Differentiating between the ACF and PACF
5. Identifying 3 types of time series models: RW/WN, Regression & ARIMA.
6. Interpreting Autoregressive models of different orders with ACF & PACF.